$$\min \quad z = 9x_1 + 12x_2$$

$$\text{s.t.} \quad -3x_1 - x_2 \leq 4$$

$$-x_1 - 2x_2 \leq 6$$

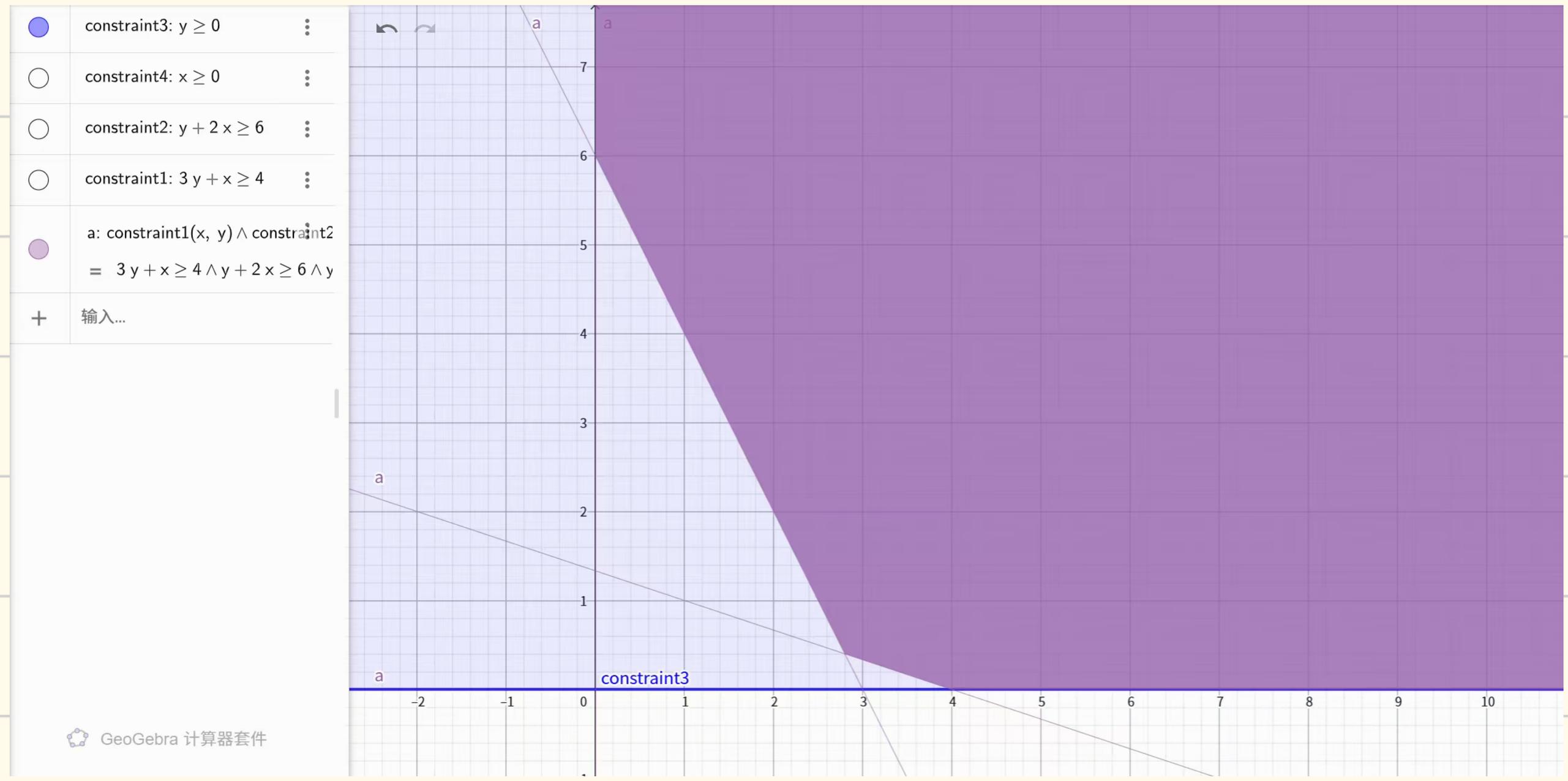
$$x_1, \quad x_2 \geq 0$$

$$\min \quad z = 9x_1 + 12x_2$$

$$\text{s.t.} \quad -3x_1 - x_2 + s_1 = 4$$

$$-x_1 - 2x_2 + s_2 = 6$$

$$x_1, \quad x_2, \quad s_1, \quad s_2 \geq 0$$



Basis	x1	x2	s1	s2	RHS
Row0	9	12	0	0	0
s1	-3	-1	1	0	-4
s2	-1	-2	0	1	-6

Basis	x1	x2	s1	s2	RHS
Row0	0	0	1 1/5	5 2/5	37 1/5
x1	1	0	- 2/5	1/5	2/5
x2	0	1	1/5	- 3/5	2 4/5

Step 1: Solve the relaxed LP Model

the optimal solution $(x^{LP,*})$: $x_1 = \frac{2}{5}$, $x_2 = \frac{14}{5} = 2\frac{4}{5}$

fractional part: $\frac{2}{5} < \frac{4}{5}$

so choose x_2 to extract the equation:

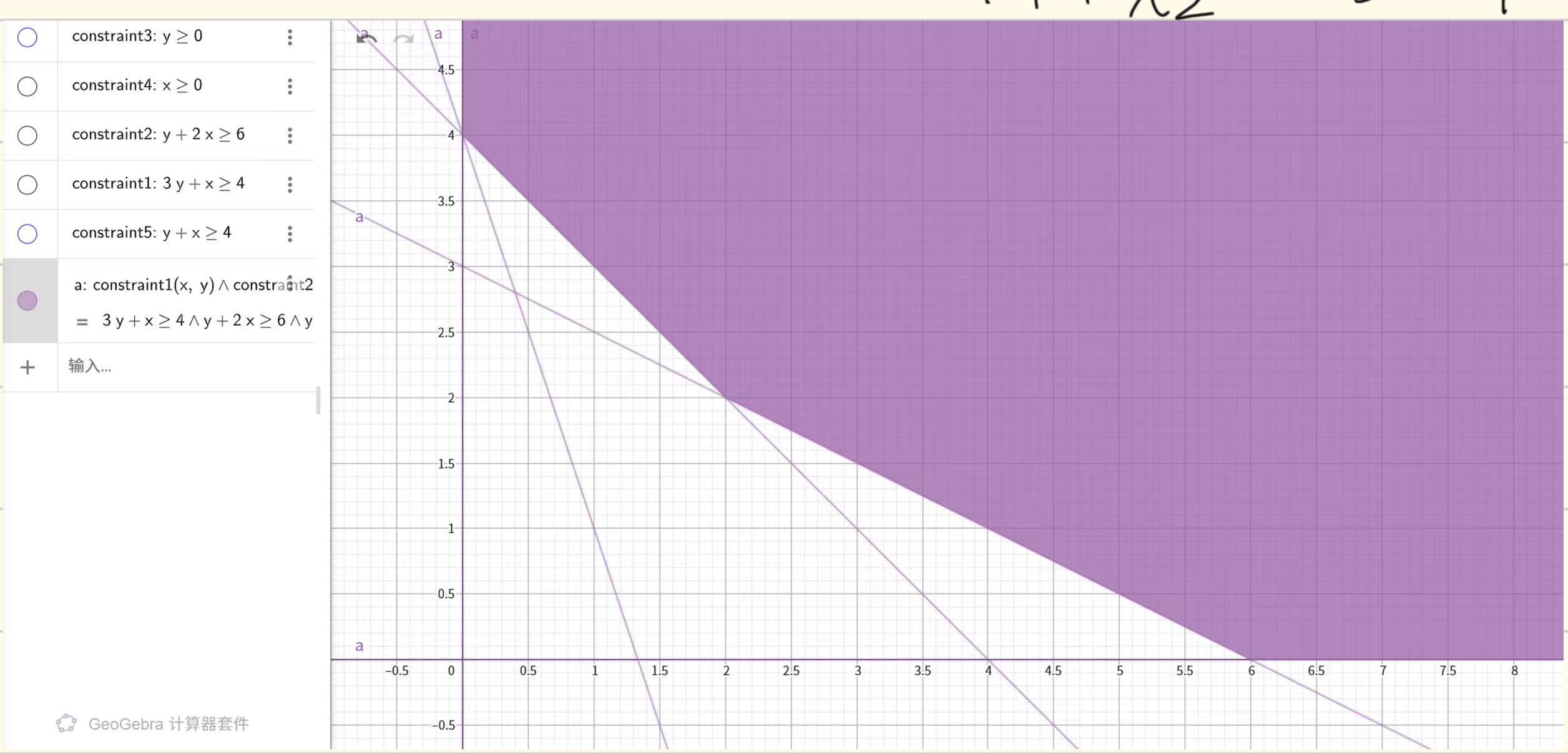
$$x_2 + \frac{1}{5}S_1 - \frac{3}{5}S_2 = \frac{14}{5}$$

$\Rightarrow x_2 - S_2 \leq 2$ Add new cutting plane

$x_2 - (x_1 + 2x_2 - 6) \leq 2$, $x_1 + x_2 \geq 4$

introduce slack variables S_3:

$$x_1 + x_2 - S_3 = 4$$



$$\min \quad Z = 9x_1 + 12x_2$$

$$\text{s.t.} \quad 3x_1 + x_2 + s_1 = 4$$

$$x_1 + 2x_2 + s_2 = 6$$

$$x_1 + x_2 - s_3 = 4$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Basis	x1	x2	s1	s2	s3	RHS
Row0	0	0	1 1/5	5 2/5	0	37 1/5
x1	1	0	- 2/5	1/5	0	2/5
x2	0	1	1/5	- 3/5	0	2 4/5
s3	1	1	0	0	-1	4

Basis	x1	x2	s1	s2	s3	RHS
Row0	0	0	3	9	3	42
x1	1	0	1	0	-1	2
x2	0	1	1/2	1	- 1/2	2
s1	0	0	1	2	-1	2

the optimal solution $(x^{LP,*})$: $x_1 = 2$, $x_2 = 2$

Both x_1 and x_2 are integer

So the optimal solution is $(x_1, x_2)^T = (2, 2)$

$\min \quad Z = 42$